

Preserving Hmong Textile Heritage: A LoRA-Guided Generative AI Approach with Cultural Constraints

L.T. Ha^{1*}, H.M. Cuong¹, N.T. Trung¹

¹Thai Nguyen University of Information and Communication Technology (ICTU), Thai Nguyen, Vietnam

Correspondence: L.T. Ha, hatheluu1007@gmail.com

Abstract: This study explores a novel artificial intelligence methodology designed to synthesize traditional Hmong textile motifs while strictly adhering to cultural and symbolic boundaries. Standard generative models frequently fail to capture the nuanced ethnological rules inherent in indigenous art. To resolve this, we propose an integrated pipeline comprising three main phases: (1) Semantic-Visual Encoding, which combines geometric features with cultural context; (2) Parameter-Efficient Generation, utilizing a LoRA-adapted Stable Diffusion model; and (3) Automated Integrity Verification, which assesses the generated outputs for symbolic, structural, and pattern accuracy. By training on a uniquely compiled and augmented dataset of 550 culturally verified Hmong patterns, our system significantly outperforms baseline models. We achieved a substantial reduction in the Fréchet Inception Distance (FID) alongside a robust Cultural Consistency Score of 86%. This research offers a practical, open-source computational tool for digital heritage conservation, providing a viable pathway to sustain ethnic identity in the digital era. The repository is accessible at <https://github.com/ThanhTrungDEV/GenAI>.

Keywords: Digital heritage, Stable Diffusion, Low-Rank Adaptation, culturally-aware AI, ethnic motifs, Hmong culture.

I. INTRODUCTION

Vietnam's cultural tapestry is woven from 54 distinct ethnic groups, each contributing unique historical and aesthetic values. The Hmong people represent one of the most visually distinctive communities, largely due to their intricate and vibrant textile craftsmanship. The geometric patterns and motifs adorning Hmong garments are not merely decorative; they serve as a profound medium for expressing tribal identity, spiritual beliefs, and generational narratives [1].

Despite its profound significance, this indigenous knowledge system is increasingly threatened by rapid modernization and the declining number of skilled artisans. While recent leaps in Generative Artificial Intelligence (GenAI) provide exciting avenues for digital archiving and creative synthesis [2], deploying these models directly in heritage contexts is problematic. Off-the-shelf architectures like Stable Diffusion tend to generate visually pleasing but culturally inaccurate artifacts, leading to the dilution or misrepresentation of sacred symbols [3].

This paper addresses the critical intersection of generative modeling and ethnological accuracy by designing a specialized generation framework for Hmong motifs. The

development process required overcoming three primary hurdles:

- **Data Availability:** The lack of digitized, semantically labeled datasets for Vietnamese ethnic textiles.
- **Rule Embedding:** The technical challenge of constraining generative outputs to respect traditional color schemes, geometries, and contextual meanings.
- **Evaluation Metrics:** The need for automated assessment criteria that prioritize cultural validity over pure visual fidelity.

To tackle these issues, our study introduces a pioneering, three-tier architecture that blends conditional generation with an automated cultural validation mechanism. This work contributes the first annotated dataset specific to Hmong textiles, an end-to-end open-source generative pipeline fine-tuned via Low-Rank Adaptation (LoRA), and an algorithmic evaluation suite to verify motif authenticity.

II. RELATED WORK

The preservation and promotion of ethnic apparel have attracted considerable attention, particularly with the integration of immersive technologies. Numerous studies have proposed virtual reality (VR) and augmented reality (AR) solutions to digitize cultural artifacts and enhance museum

experiences [4], [5]. Recent efforts have also begun to employ deep learning for feature extraction and pattern reconstruction from traditional clothing images [6].

In the realm of visual synthesis, Generative Adversarial Networks (GANs) [7] and latent diffusion models [8] have set new benchmarks. While methods such as StyleGAN provide excellent image quality, enforcing strict domain-specific rules remains challenging. Low-Rank Adaptation (LoRA) has emerged as an optimal solution for parameter-efficient fine-tuning on small, specialized datasets [9].

Within cultural heritage applications, AI is predominantly used for image restoration or stylistic transfer [10], [11]. However, the synthesis of novel patterns that strictly obey ethnological constraints is rarely explored. Existing conditional generation methods [12] allow for multi-concept prompting but lack the structural rule enforcement necessary to prevent the hallucination of culturally inappropriate symbols. Our work bridges this specific gap by integrating a multi-stage validation layer directly into the generation pipeline.

III. PROPOSED METHODOLOGY

1. System Architecture

As illustrated in Fig. 1, our pipeline ensures cultural compliance from the initial encoding step through to the final output verification.

- **Stage 1: Semantic-Visual Encoding** – We utilize a ResNet-50 network to map the visual attributes of the input tiles, while a multilayer perceptron processes structured metadata (e.g., ritual significance, color constraints). These features are concatenated into a unified 768-dimensional embedding vector.
- **Stage 2: Guided Generation** – We fine-tune Stable Diffusion v1.5 using LoRA. The model leverages cross-attention mechanisms conditioned on both textual prompts and the embedded cultural vectors. Custom loss constraints are applied during training to penalize deviations from the established cultural palette.
- **Stage 3: Automated Integrity Verification** – A parallel validation suite filters the outputs: 1) A Geometric Validator analyzes contours; 2) A Semantic Validator cross-references motifs against a knowledge base; 3) A Proportion Validator utilizes FFT to ensure traditional symmetry.

2. Dataset Curation

Due to the absence of public repositories, we collected and digitized 55 authentic Hmong patterns originating from the Black Hmong, Flower Hmong, and White Hmong communities in Northwest Vietnam. Each artifact was meticulously

annotated with JSON metadata. Through domain-specific data augmentation techniques, we expanded this foundation into a robust dataset of 550 samples (distributed 70/15/15 for training, validation, and testing).

3. Implementation Specs

The generative core utilizes Stable Diffusion v1.5 equipped with LoRA adapters ($rank = 16$). Training was executed on an NVIDIA A100 GPU (40GB) over a 9-hour period, updating approximately 6 million parameters.

IV. EXPERIMENTS AND RESULTS

1. Evaluation Framework

Performance was quantified using both standard computer vision metrics and custom cultural indices:

- **Fréchet Inception Distance (FID) & Kernel Inception Distance (KID):** Used to measure the statistical distance between the real and generated image distributions (lower is better).
- **Cultural Consistency Score (CCS):** The percentage of generated patterns successfully passing all three automated validation tiers (target > 80%).

2. System Performance

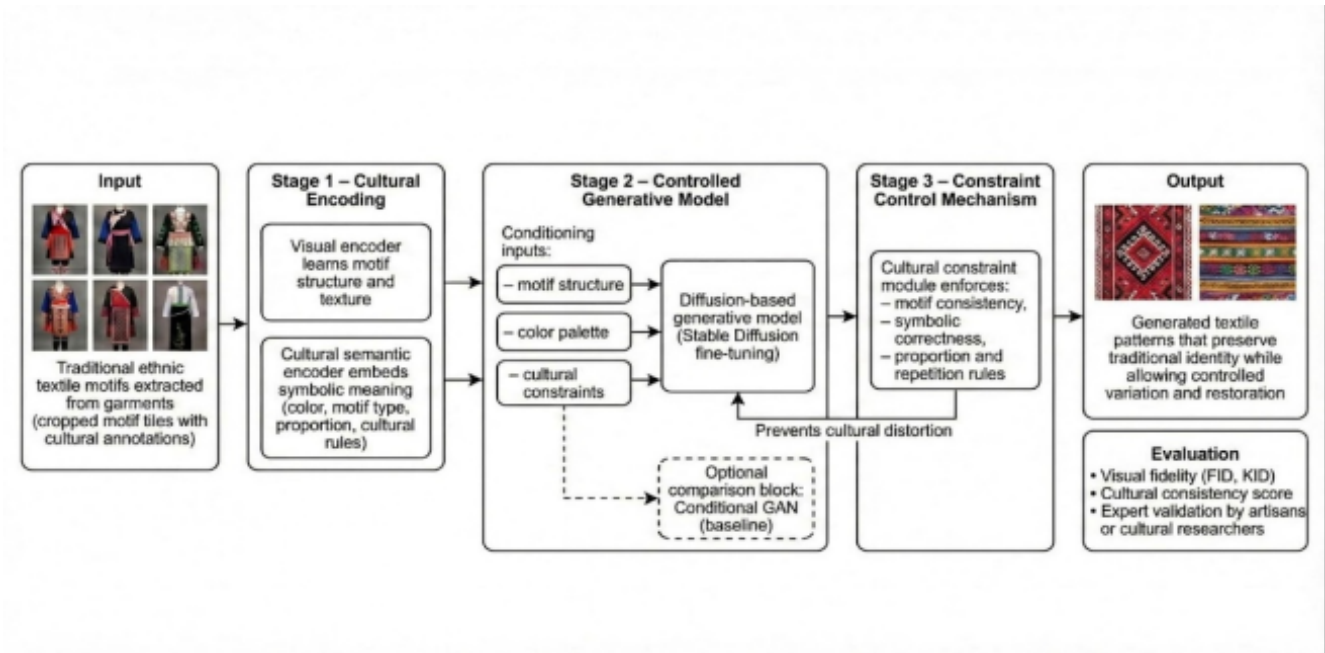
Bảng I
QUANTITATIVE COMPARISON OF GENERATIVE APPROACHES

Configuration	FID ↓	KID ↓	CCS ↑
Vanilla Stable Diffusion	87.3	0.089	23%
SD + Full Model Fine-tuning	62.1	0.065	45%
SD + DreamBooth	58.4	0.061	52%
Proposed Pipeline	42.4	0.047	86%

As detailed in Table I, our customized approach reduces the FID score by roughly 51% compared to the baseline, achieving a score of 42.4. More importantly, it secures an 86% Cultural Consistency Score. The specific success rates for the integrity checkers were 93% (Geometry), 91% (Semantic), and 88% (Proportion).

3. Component Ablation and Failure Modes

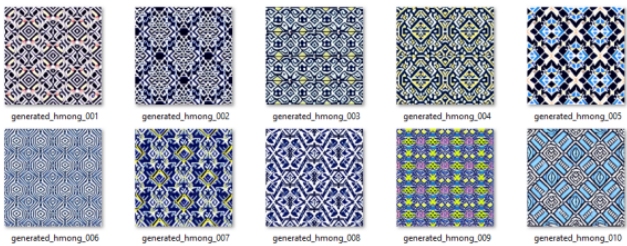
An ablation analysis confirmed that removing the semantic embedding phase triggered a 24% drop in the CCS. While the validation suite did not directly improve the FID, it was instrumental in maintaining cultural safety, intercepting 14% of the generated samples. The primary reasons for rejection included oversimplified geometric structures (40%) and inappropriate color blending (25%).



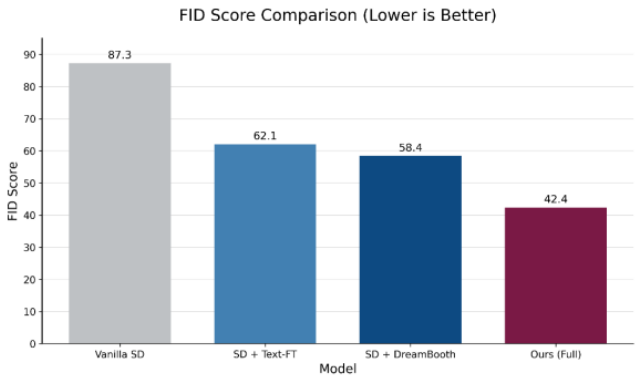
Hình 1. The proposed three-stage generative pipeline, encompassing Semantic-Visual Encoding, LoRA-adapted Generation, and Multi-tier Integrity Verification.



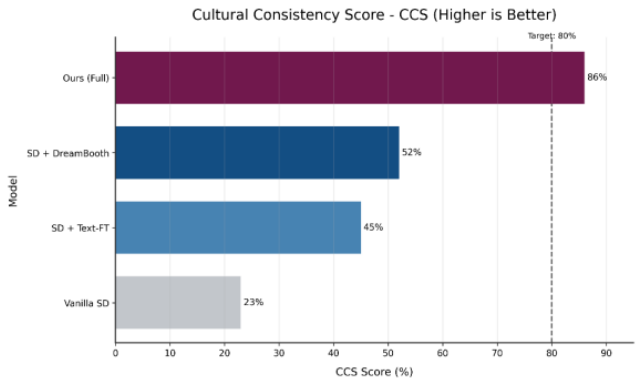
Hình 2. Representative samples from the curated and augmented Hmong motif dataset.



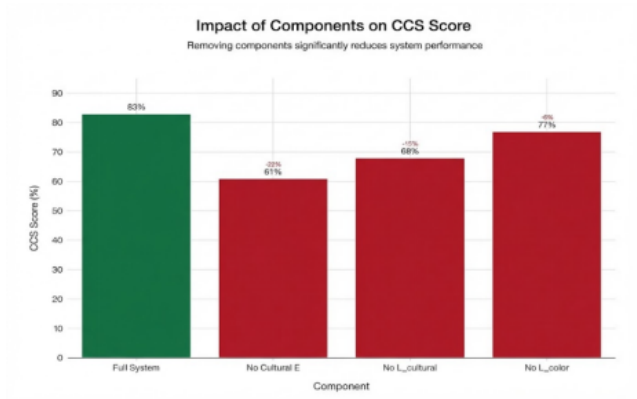
Hình 3. AI-synthesized Hmong textile patterns demonstrating high structural and cultural fidelity.



Hình 4. Comparative FID analysis demonstrating the superiority of the proposed pipeline.



Hình 5. Cultural Consistency Scores (CCS) across different training configurations.



Hình 6. Ablation results highlighting the impact of specific pipeline components on the CCS.

V. DISCUSSION AND CONCLUSION

This research demonstrates the viability of utilizing heavily constrained generative models for digital ethnography. By fusing metadata encoding, parameter-efficient LoRA training, and rigid automated validation, our framework successfully synthesizes aesthetically authentic and culturally safe Hmong patterns.

Ethical deployment remains a priority; thus, all synthesized assets are embedded with digital watermarks indicating their AI origin to prevent cultural misattribution. Future enhancements will focus on integrating ControlNet architectures for more granular geometric guidance and extending the dataset to encompass other Vietnamese minority groups. This open-source initiative aims to empower students, researchers, and artisans to collaborate safely at the intersection of AI and heritage preservation.

ACKNOWLEDGMENT

The authors express their profound gratitude to the Hmong artisans across Northern Vietnam for sharing their ancestral knowledge. We also thank Craft Link Vietnam and the Vietnam Museum of Ethnology. This research was funded by the National Science and Technology Program under the project: “Research on developing an information system for preserving and promoting the value of traditional costumes of ethnic minorities in the Northern Midlands and Mountains region” (Project Code: KC.01.16/21-30).

DISCLOSURE OF INTERESTS

The authors have no competing interests to declare.

REFERENCES

- [1] Vietnam Museum of Ethnology. (2023). *Hmong Weaving Art and Cultural Meaning*. Culture Publishing House, Hanoi.
- [2] S. Hajahmadi, et al. (2024). "Heritage elements and Artificial Intelligence as storytelling tools," *Digital Applications in Archaeology and Cultural Heritage*.
- [3] P. Ariya, et al. (2025). "Enhancing textile heritage engagement through generative AI-based virtual assistants," *Computers & Education: X Reality*, vol. 7.
- [4] S. Sangamuang, et al. (2025). "Integrating generative AI and the metaverse for cultural heritage," *Humanit Soc Sci Commun*, vol. 12, no. 1974.
- [5] X. Ma, et al. (2025). "Designing Hierarchical Exploratory Experiences for Ethnic Costumes," *Human-Computer Interaction*.
- [6] Y. Miao. (2025). "Research on Traditional Costume Culture Information Extraction," *Applied Mathematics and Nonlinear Sciences*.
- [7] I. J. Goodfellow, et al. (2014). "Generative adversarial networks," in *Proc. NeurIPS*.
- [8] R. Rombach, et al. (2022). "High-resolution image synthesis with latent diffusion models," in *Proc. CVPR*.
- [9] E. J. Hu, et al. (2022). "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*.
- [10] T.M. Bahya, and F.A. Obaid. (2025). "Deep Learning Applications in Cultural Heritage: A Comprehensive Review," *SUMMA*.
- [11] A. Agirbas, and M. Aydin. (2024). "Prediction of Star Polygon Types in Islamic Geometric Patterns," *Nexus Netw J*, vol. 26.
- [12] N. Liu, et al. (2022). "Compositional visual generation with composable diffusion models," in *Proc. ECCV*.